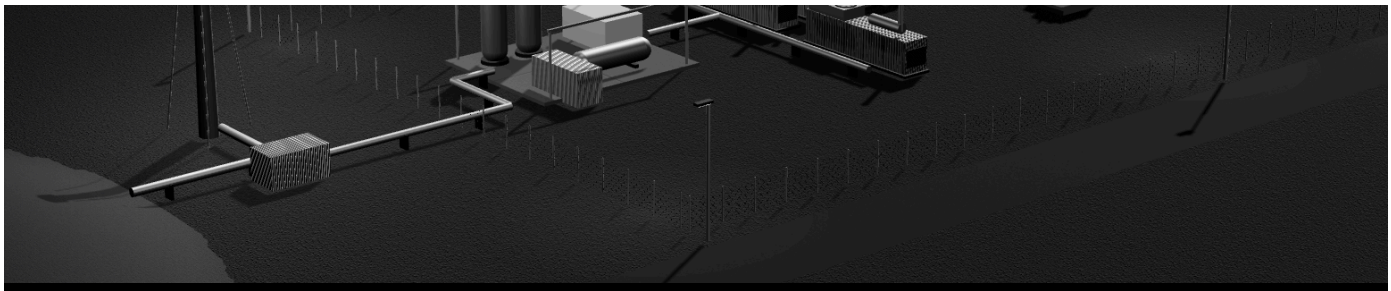
Cherokee Run Landfill, Bellefontaine, Ohio: a first sizing

What the gas Cherokee Run now flares could power, how much land it would take, and what it could earn each year.

6.2 MW	6	12 ac	564k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
$846k to $1.41M	$12k to $25k	$859k to $1.44M



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=2214&CH4=50

THE MATH

Landfill gas collected (2023 average)	2,214 scfm
Methane content (2023)	50.4%
Energy in the gas	67.8 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 6.2 MW
Compute (power less cooling and losses, /1.2)	about 5.1 MW
Compute modules (about 1 MW each)	6
Gas engines (running plus one spare)	5 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 12 acres
Gas bought per year (95% run time)	564k MMBtu
Gas payment at $1.50 to $2.50 per MMBtu	$846k to $1.41M
Land lease at $1,000 to $2,000 per acre per year	$12k to $25k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed or floored price.

WORTH CHECKING

- Who holds the gas rights since the 2020 engine shutdown
- Whether the old 12 kV point and pad can be reused
- Latest flow and methane data, and the outlook to closure (2063)
- Rushcreek Township zoning for the parcel
- The old engine plant ran 4.8 MW; engine choice sets the final figure

Source: EPA Greenhouse Gas Reporting Program, Subpart HH, facility 1002480, 2023 report (gas destroyed on site since 2020); EPA LMOP database, 2024 release. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.

Renzo Bagatoli, CEO · renzo@accretioncompute.com